

Molecular Dynamics of Mutated A β 16-21 Amyloid Aggregate

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Abstract

This study investigates the effects of a F19E and F20E mutation on the molecular dynamics of the A β 16-21 amyloid-beta fragment. For both the wild-type and mutant, four aggregated peptides were solvated with explicit water and NaCl, then observed over time. We quantified these observations through RMSD, DSSP, SASA, hydrogen bond, and radius of gyration calculations. To elucidate the exact properties that cause this change, we predicted aggregation propensity of other mutants to isolate the varied physicochemical characteristic. Overall, we found that the F19E/F20E mutation decreased the stability of the four-peptide aggregate and resulted in greater aqueous interactions than the wild type amyloid-beta.

Introduction

Alzheimer's disease (AD) is a neurodegenerative disorder characterized by a gradual decline in cognitive function and onset of dementia. AD is hypothesized to be caused by the aggregation of two misfolded proteins in the brain: amyloid beta (A β) plaques and tau. According to the amyloid cascade hypothesis, A β accumulation in the form of fibrils is involved in the pathogenesis of AD. Researchers have discovered that a primary driver of aggregation is two phenylalanine (FF) residues, causing aggregation through hydrophobicity.

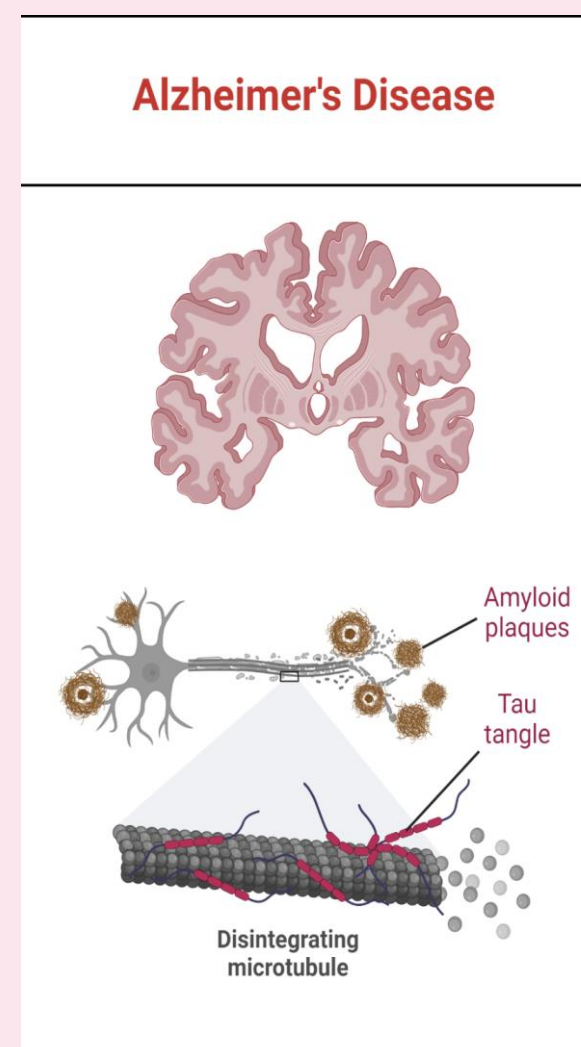


Fig. 1: Alzheimer's pathology. Adapted from brightfocus.org published literature.

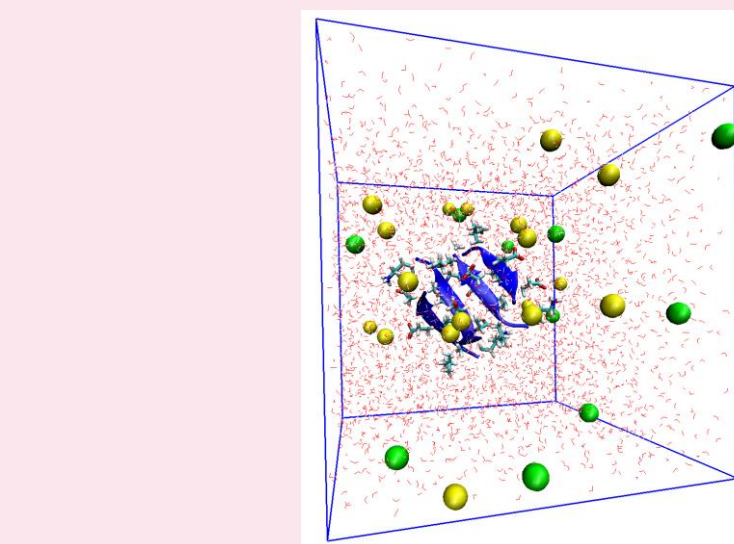


Fig. 2: Mutant Amyloid Beta Snapshot from VMD

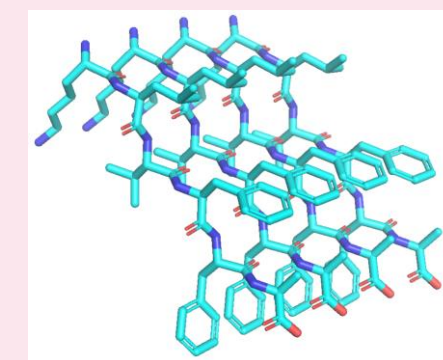


Fig. 3: Wild type aggregate

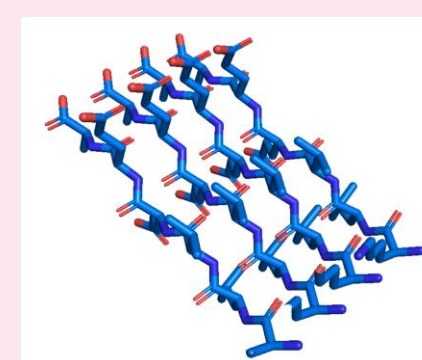


Fig 4: Mutant aggregate

Mutation	HSA	NHSA	a4vAHS
KLVFFA	5.208	0.868	0.810
KLWWA	3.703	0.617	0.571
KLVYYA	3.959	0.660	0.612
KLVLAA	4.423	0.737	0.686
KLQQQA	all 0.000 except Q = 0.115	0	all 0 except Q = 0.095
KLVEEA	all 0.000 except E = 0.043	0	all 0 except E = 0.023
KLKKKA	all 0 except V and K = 0.291	0	all 0 except V and K = 0.126

Table 1: Predicted aggregation statistics for different KLVFFA mutants

Results

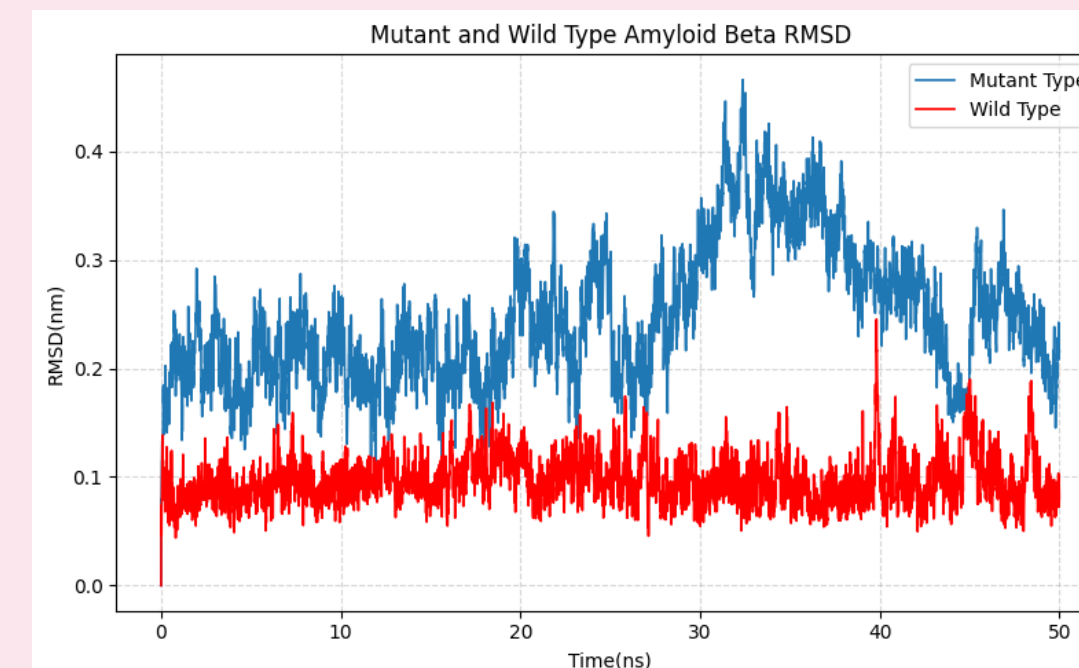


Fig. 5: Mutant and Wild-type RMSD analysis

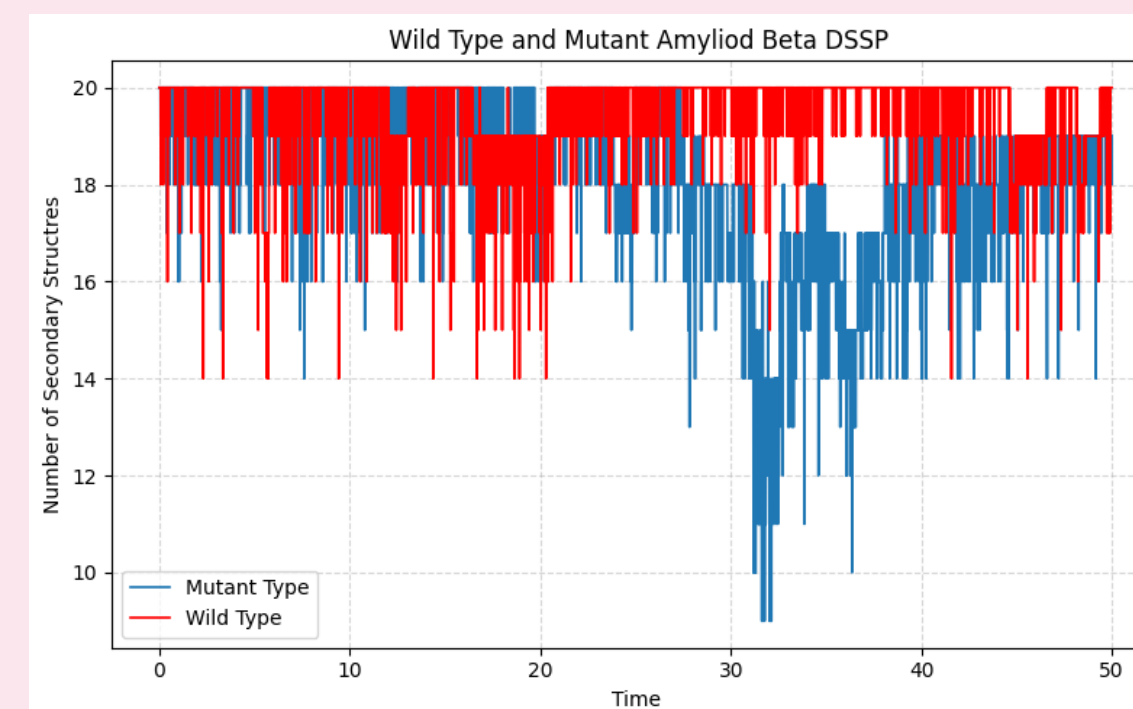


Fig. 7: Mutant and Wild-type DSSP

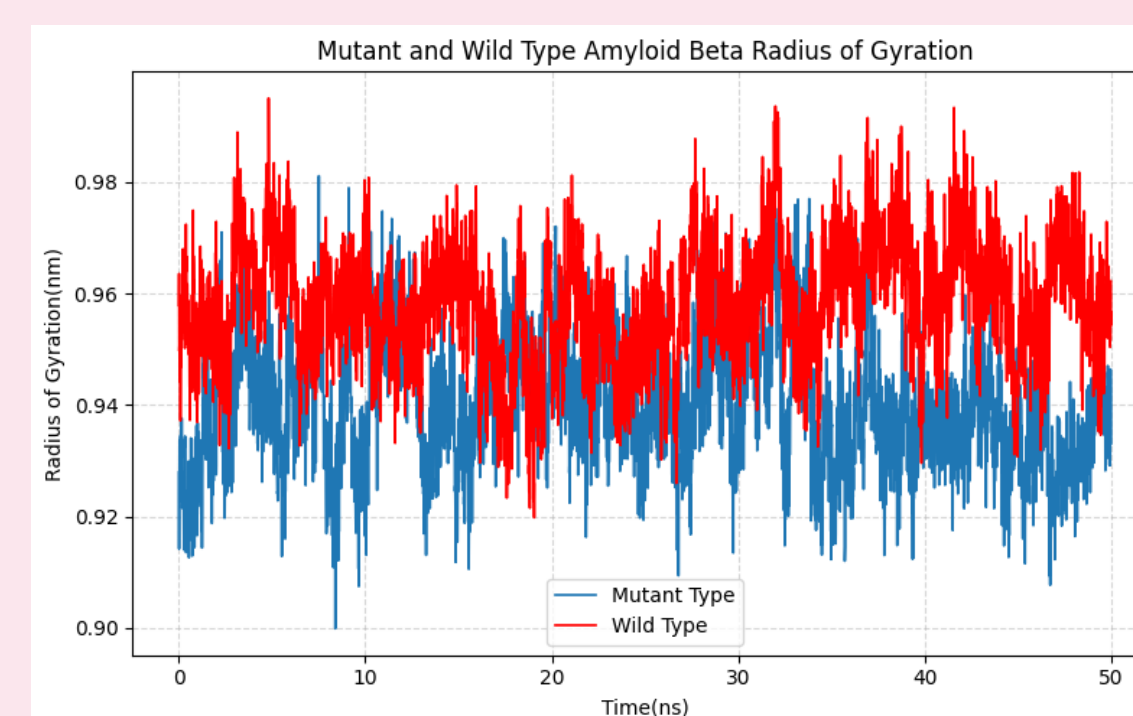


Fig. 9: : Mutant and Wild-type radius of gyration

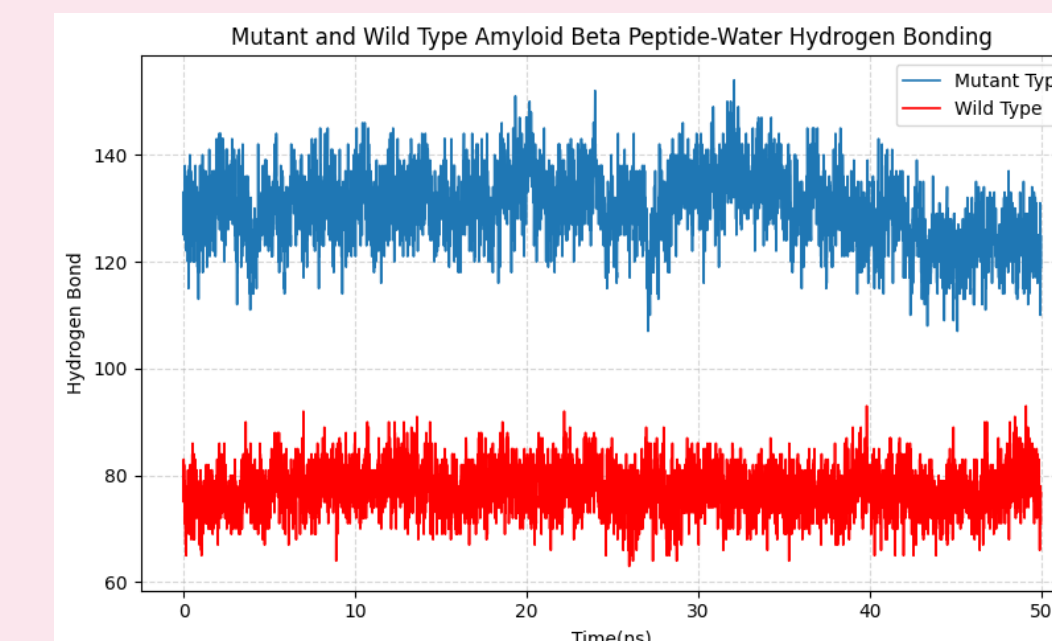


Fig. 6: Mutated and Wild-type peptide-water hydrogen bonds

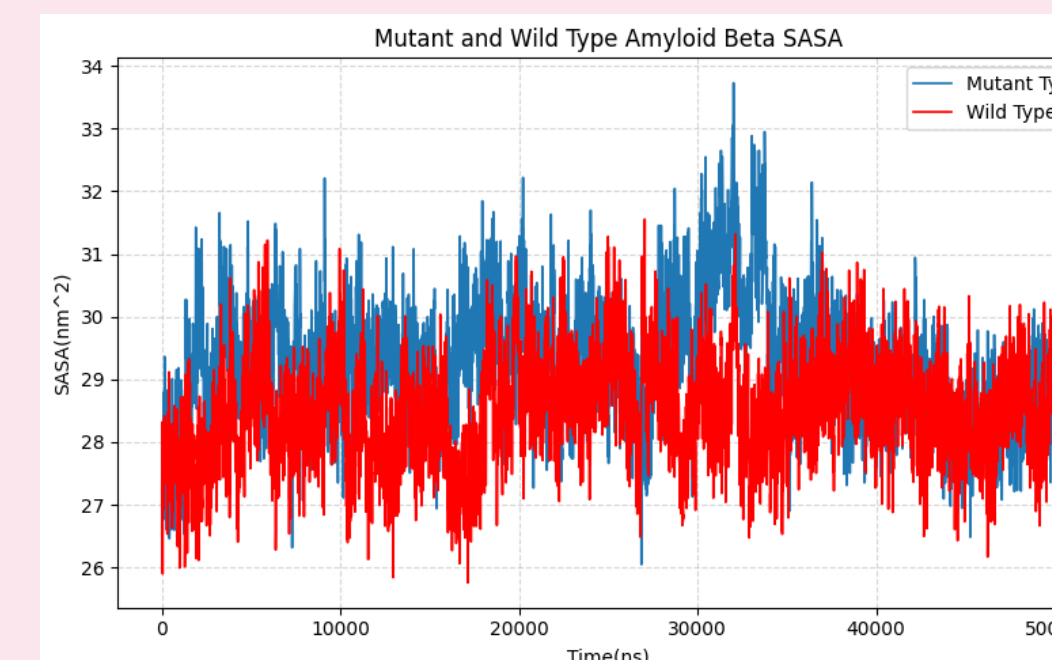


Fig. 8: Mutant and Wild-Type SASA

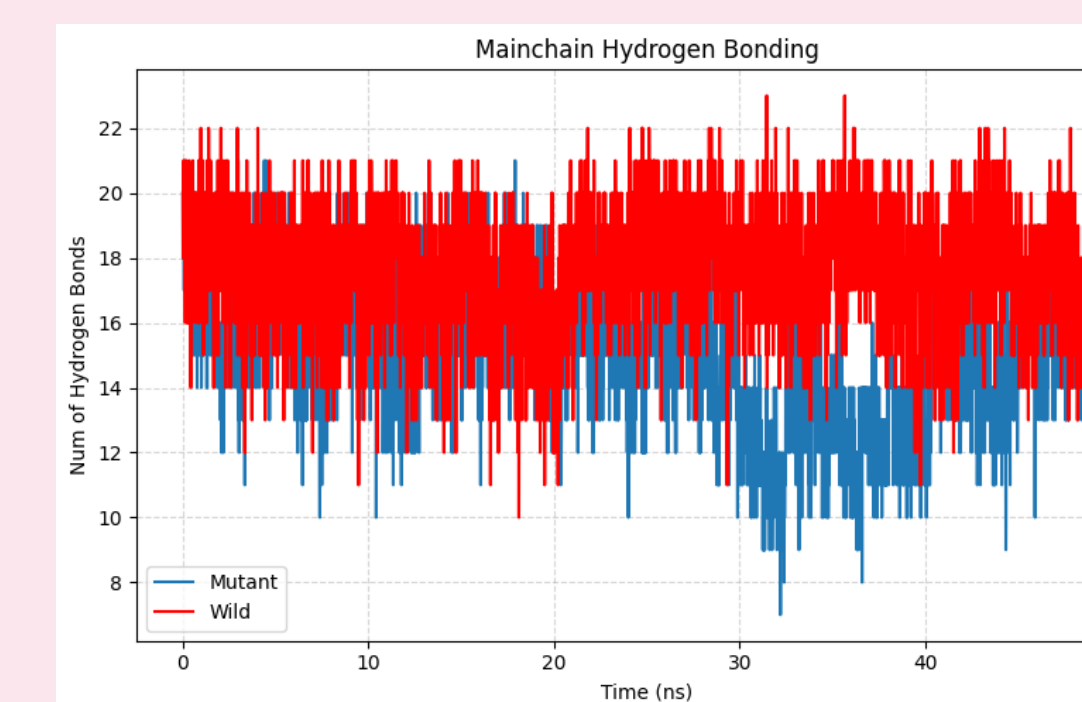


Fig. 10: Mutant and Wild-type main chains

Conclusion

The results suggest that the F19E/F20E mutation substantially changes the behavior of the A β 16-21 peptide aggregate compared to the wild type. Greater RMSD fluctuations suggest the mutant structure undergoes significant structural changes, while the DDSP plot indicates a highly varied secondary structure. Lower radius of gyration suggests that the mutant is adapting a more compact conformational ensemble, and the increased peptide-water hydrogen bonding is consistent with the replacement of hydrophobic phenylalanine residues by charged glutamate residues, resulting in stronger interactions with the aqueous environment. Together, these findings suggest that the F19E/F20E substitution decreases the β -sheet stability and compactness of the aggregate and increases interactions with the aqueous environment. The aggregation propensity prediction found that overall positive and negative charges decreased aggregation hot spot areas to approximately zero, while tyrosine produced the greatest decrease. These results suggest that larger amino acids may produce steric hindrance that makes aggregation or burial unfavorable.

Limitations & Future Research

A core limitation in this study is the short duration of the simulation, therefore the scope of the simulation may not include significant changes in the protein. The force fields used are also empirical approximations and may simplify true quantum electron behavior. We hope that this research can be applied to improve the development of therapeutics targeting amyloid beta fibrils in Alzheimer's patients.

Acknowledgements

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References



Methods

Molecular Dynamics Simulation

- Structure Prediction** $\rightarrow$ Obtained predictions of the wild-type peptide sequence KLVFFAE and of the mutant peptide KLVEEAE using AlphaFold
- System preparation** $\rightarrow$ Predicted structures were prepared in Google Colab using GROMACS with the AMBER99SD-ILDN force field. Topologies, bonding parameters and the total system charge were generated
- Solvation and Ion Addition** $\rightarrow$ Built the water box and solvated it using water model SPC 126. Ca²⁺ and Na⁺ ions were added according to the selected ionic conditions
- Equilibration** $\rightarrow$ The systems were relaxed and equilibrated through NVT and NPT equilibration
- Production MD** $\rightarrow$ Production molecular dynamics simulations were performed to generate trajectories for the wild-type and mutant peptides

Trajectory and Structural Analysis

- Trajectory Processing** $\rightarrow$ Removed Periodic-boundary (PBC) visualization artifacts to make molecules whole and keep the peptide centered in the box.
- Backbone RMSD** $\rightarrow$ Backbone root-mean-square deviation (RMSD) was calculated to assess structural deviation and stability over the simulation.
- Radius of Gyration** $\rightarrow$ Calculated the radius of gyration to find the mass-weighted spatial distribution of the peptide around its center of mass.
- Secondary Structure** $\rightarrow$ DSSP analysis was applied to the trajectories to characterize changes in secondary structures (alpha helices, beta strands and loops)
- Hydrogen-Bond Analysis** $\rightarrow$ The number of hydrogen bonds was calculated over the trajectory to compare hydrogen-bonding behavior
- SASA** $\rightarrow$ Evaluate the peptide interaction with its aqueous environment

Physicochemical analysis

- Predict aggregation statistics** for different mutated sequences using the AGGRESCAN server
- Input FASTA sequences** of each mutant and obtain HSA (hot spot area), NHSA (normalized hot spot area), and a4vAHS (average a4v per hot spot) to describe the aggregation propensity.